

MAT 426, Class 5: Implementing GEPP

Example 1. Let A be an $n \times n$ matrix.

(a) If $n = 4$, how do we find the L and P in the $PA = LU$ factorization?

(b) How do we find L and P in the general case?

Question: How do we implement the $PA = LU$ factorization?

Example 2. Consider the matrix $A = \begin{bmatrix} 2 & 2 & 2 & 1 \\ 4 & 4 & 7 & 3 \\ -2 & 4 & -1 & 2 \\ 6 & 9 & 5 & 8 \end{bmatrix}$

(a) Compute the $PA = LU$ factorization.

- (b) How can we determine the entries in L and P of the $PA = LU$ factorization as we go?
- Store the _____ of the multipliers in the _____ triangle of A as before.
 - At step k , we swap rows _____ and _____ (which contains the entry in column k that is largest in absolute value) in the matrix A . For L , swap rows k and r_k for the _____ that have already been computed; i.e. columns 1 to $k - 1$.
 - For P , we store the _____ that are interchanged at each step in a _____.
- (c) Repeat (a), storing everything in A along with a vector containing the row interchanges. What are the matrices in the $PA = LU$ factorization?

LU FACTORIZATION USING GEPP

Input:

Outputs:

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For $k = 1, 2, \dots, n - 1$ do

1. (Find the pivot row). Find r_k such that $|a_{r_k,k}| = \max_{k \leq i \leq n} |a_{i,k}|$. If $a_{r_k,k} = 0$, stop. Otherwise, continue.
2. (Interchange rows r_k and k). $a_{kj} \longleftrightarrow a_{r_k,j} \quad j = 1, \dots, n$.
3. (Form the multipliers) $a_{i,k} \equiv m_{i,k} = \frac{a_{i,k}}{a_{k,k}} \quad (i = k + 1, \dots, n)$
4. (Update the entries) $a_{i,j} \equiv a_{i,j} - m_{i,k}a_{k,j} = a_{i,j} - a_{i,k}a_{k,j} \quad i = k + 1, \dots, n; j = k + 1, \dots, n$